

(a) Hazard identification

once per episode · no simulator state

Instruction ℓ

“put the mug...”

RGB-D views

+ proprioception

Object names

symbolic list only



VLM predicates

Referenced(x)
Protected(x) *Hot(x)*

3 votes · cached · API-free

Geometric

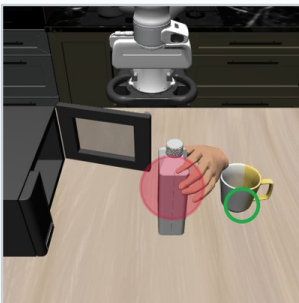
NearestToPath(x)
Moving(x)

detector + depth

Three-literal election rule

$\text{Hazard}(x) := \text{Protected}(x) \vee \text{Moving}(x) \vee$
 $(\text{Pred}(x) \wedge \neg \text{Mentioned}(x) \wedge \neg \text{Target}(x))$
abstains when no object is elected

Elected guard set



bottle-in-hand → guarded
mug (target) → exempt

(b) Routing

three regimes

Avoid

geometric hazard
off the task path

Disengage

$\text{Protected} \wedge \text{Referenced}$
handover: keep-out off

Refuse

the instruction is the
hazard → no execution

one predicate layer,
three altitudes

(c) Noise-space enforcement

every replan · frozen policy · API-free

$z_1 \dots z_K \sim N(0, I)$
K = 8 seeds

Frozen $\pi_{0.5}$ flow map $F(z, \text{obs})$

10 Euler steps · shared vision-language KV cache

Select

best-of-K

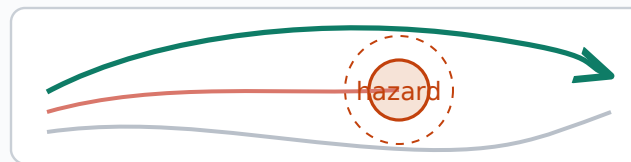
Tilt

FK reweighting

Ascend

∇_z prefix margin

search over the safe preimage $S = \{z : M(F(z)) \geq 0\}$
+ soft repulsor on the velocity target (obstacle-on-grasp)



(d) Certificate

checked on output

DCBF chain

$B_j \geq (1-\gamma) B_{j-1}$

re-checked on the
executed 5-step prefix
of the decoded chunk
mechanism-independent

**Execute certified
prefix (5 actions)**

all K infeasible →
argmax margin runs,
true margin reported
(fail-visible)

► VLM answers local predicates; the logic decides

► actions are always $F(z^*)$: exactly on the policy manifold

replan every 5 control steps · fresh observation · $\text{Moving}(x)$ re-read · guard set persists